Problem analysis – DANUBE4all

PLEASE NOTE: This document presents the individual problems considered in the problem analysis of the Danube River basin under the DANUBE4all project. Most descriptions are drawn from the DanubeSediment_Q2 project report "Hydromorphological Survey and Assessment in the Framework of JDS5 and DanubeSediment_Q2 – Operational Manual." For detailed guidance on parameter assessments, please refer to the aforementioned report.

Problem criteria:
No = This **problem is not present** in the reach
Yes = This **problem is present** in the reach

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1. **Change in frequency of flood flow (C 1.1)**

   Change of occurrence of a flood flow with a 2-year return period (HQ2) is assessed by comparing the present with a past period (most historical data; if possible, past data should reflect time before anthropogenic measures affecting flood flow were implemented). The occurrence of a flood flow with a 2-year return period (HQ2) might be changed due to anthropogenically induced alterations of flows in the catchment upstream the reach (i.e. water abstractions, dams retaining and buffering water) as well as climate change related impacts.

***Problem criteria* - Change in frequency of flood flow (C 1.1)**	
No	No or slight changes in frequency of occurrence (±≤ 10%) of a HQ_2-discharge within present and past period are assessed.
Yes	Changes in frequency of occurrence (±> 10%) of a HQ_2-discharge within present and past period are assessed.

2. **Change in mean flow (C 1.2)**

   Change in mean monthly flows (MQ) is assessed by comparing present versus past period (most historical data; if possible, past data should reflect time before anthropogenic measures affecting mean flow were implemented). Thereby, the periods should be ≥ 20 years and homogeneous, meaning no relevant anthropogenic impacts on mean flow occurred within the periods (homogeneity is more important than length of period; still sufficient length of period is necessary). is necessary). Mean monthly flows (MQ) might be changed due to anthropogenically induced alterations of flows in the catchment upstream the reach (i.e. water abstractions, dams retaining and buffering water etc.) as well as climate change related impacts.

***Problem criteria* - Change in mean flow (C 1.2)**	
No	No or slight changes in frequency of occurrence (±≤ 10%) of a HQ_2-discharge within present and past period are assessed.
Yes	Changes in frequency of occurrence (±> 10%) of a HQ_2-discharge within present and past period are assessed.

3. **Change in minimum flow (C 1.3)**

   Change in minimum monthly flows ($NQ_{monthly}$) comparing two time periods of ≥ 20 years is assessed. Thereby, the present period vs. past period (most historical data; if possible, past data should reflect time before anthropogenic measures affecting minimum flow were implemented) is to be assessed, where the two periods are to be homogeneous, meaning no relevant anthropogenic impacts on minimum flow occurred within the periods (homogeneity is more important than length of period; still sufficient length of period is necessary). Minimum monthly flows (NQmonthly) might be changed due to anthropogenically induced alterations of flows in the catchment upstream the reach (i.e. water abstractions, dams retaining and buffering water etc.) as well as climate change related impacts.

***Problem criteria* - Change in minimum flow (C 1.3)**	
No	No or slight changes (±≤ 10%) in minimum monthly flows ($NQ_{monthly}$) in the long term present vs. past period.
Yes	Changes (±> 10%) in minimum monthly flows ($NQ_{monthly}$) in the long term present vs. past period.

4. Change in seasonal flow (C 1.4)

Shift in timing of annual peak values of daily mean flows comparing two time periods of ≥ 20 years is assessed. Thereby, the present period vs. past period (most historical data; if possible, past data should reflect time before anthropogenic measures affecting seasonal flow were implemented) is to be assessed, where the two periods are to be homogeneous, meaning no relevant anthropogenic impacts on seasonal flow occurred within the periods (homogeneity is more important than length of period; still sufficient length of period is necessary). Change in timing of annual peak values of daily mean flows might be changed due to anthropogenically induced alterations of flows in the catchment upstream the reach (i.e. water abstractions, dams retaining and buffering water) as well as climate change related impacts.

***Problem criteria* - Change in seasonal flow (C 1.4)**	
No	No or first evidence of a change (±< 20 days) in timing of annual peak flow comparing present vs. past period.
Yes	Change (±≥ 20 days) in timing of annual peak flow comparing present vs. past period.

5. Change in bedload volume (C 2.1)

<u>C 2.1 A: Annual bedload transport</u>

Change in mean annual bedload transport (m^3 $year^{-1}$ or t $year^{-1}$) under present conditions compared to data of past mean annual bedload transport (most historical data) is assessed. If possible, past data should be based on a period rather than few years only. Bedload transport might be influenced by damming, river engineering measures, land-use change, capital dredging, etc. within the catchment upstream of the assessed reach. Sediment feeding and capital dredging in an upstream reach is here indirectly included by accounting for increase/decline in bedload supply (improvement/worsening of scores in the assessment table).

***Problem criteria* - Change in bedload volume (C 2.1 A)**	
No	No or slightly modified sediment input: mean annual bedload supply ≥ 80% of past mean annual bedload supply;
Yes	Modified sediment input: mean annual bedload supply < 80% of past mean annual bedload supply;

<u>C 2.1 B: Bedload connectivity</u>

Change in bedload supply is assessed by evaluating the presence of dams and/or sediment barriers with high trapping effect and their affecting proportion (%) of upstream catchment area. Dams and other transversal barriers can disrupt sediment connectivity, acting as bedload traps upstream and thus impact the bedload supply into a respective reach. Here, dams with significant and permanent (applies also if dam is capable of sediment release) sediment trapping effect, and sediment barriers with high trapping effect are evaluated as they are related to sediment connectivity within the catchment. Sediment barriers with high trapping effect might be small dams (not being delineation criteria) or abstraction weirs/sluicing structures or retention weirs.

***Problem criteria* - Change in bedload volume (C 2.1B)**	
No	The dam's affecting catchment area is ≤ 35%, **or** the barrier's affecting catchment area is ≤ 65% of the total catchment area upstream the assessed reach (Fig. B).
Yes	The dam's affecting catchment area is > 35%, **or**

	the barrier's affecting catchment area is > 65% of the total catchment area upstream the assessed reach (Fig. B).

6. Change in bedload grain size (C 2.2)

Change in median bedload grain sizes (D50), coarsening/fining, comparing the average D50 of two time periods of ≥ 5 years is assessed. Thereby, the present period vs. past period (most historical data, possibly before implementations of HPPs) is to be assessed, where the two periods are to be homogeneous, meaning no relevant anthropogenic impacts on bedload occurred within the periods (homogeneity is more important than length of period; still sufficient length of period is necessary). In general, a change of the median grain size (D50) within the grain size distribution could indicate an imbalance between sediment supply and sediment transport capacity. Change in bedload grain sizes, e.g., shift in grain size distribution (D50) towards coarsening could be an expected response change affected by upstream sediment barriers (e.g. reservoirs), increased sediment transport capacity, etc. What also can happen is that the bedload is much finer because it is transported over an armoured surface without exchange with the bed.

***Problem criteria* - Change in bedload grain size (C 2.2)**	
No	No or slight change in bedload D_{50} (±≤ 20%) compared to past conditions.
Yes	Change in bedload D_{50} (±> 20%) compared to past conditions.

7. Change in suspended sediment load (C 2.3)

Change in mean annual suspended sediment load (t year-1) comparing two time periods of ≥ 20 years is assessed. Thereby, the present period vs. past period (most historical data, possibly before implementations of HPPs) is to be assessed, where the two periods are to be homogeneous, meaning no relevant anthropogenic impacts on suspended sediment load occurred within the periods (homogeneity is more important than length of period; still sufficient length of period is necessary). Suspended sediment budget can significantly impact type-specific grain-size compositions, clogging processes, fine material deposits in the foreshore, silting of floodplain waters, turbidity and, as a consequence, oxygen concentrations etc.

***Problem criteria* - Change in suspended sediment load (C 2.3)**	
No	No or slightly modified suspended sediment input: deviation of annual suspended sediment supply ≤± 20% compared to past suspended sediment supply;
Yes	Modified suspended sediment input: deviation of annual suspended sediment supply >± 20% compared to past suspended sediment supply;

8. Change in suspended sediment temporal distribution (C 2.4)

This parameter assesses the temporal distribution of suspended sediment by comparing two time periods of same length ≥ 20 years. Thereby, the period after dam construction vs. the period before dam construction (most historical data) are to be assessed, where the two periods are to be homogeneous, meaning no relevant anthropogenic (land use or floodplain changes) or natural (flood event with significant suspended sediment transport) impacts on suspended sediment load occurred within the periods (homogeneity is more important than length of period; still sufficient length of period is necessary). The transport of suspended

sediments generally relies on water discharge and its velocity. Hydropower dams alter the water flow conditions as their reservoirs exhibit reduced flow velocities. Consequently, sediments settle in the reservoirs and transport of suspended sediments is changed/interrupted. During flood events opening of gates cause remobilization of suspended sediments. This anthropogenic impact alters the temporal distribution of suspended sediment transport.

***Problem criteria* - Change in suspended sediment temporal distribution (C 2.4)**	
No	The temporal distribution of suspended sediment transport is not or negligible **or** the time to cumulate 75% of suspended sediment transport is ±≤ 20% compared to its initial time (e.g., reduction from 20% to 16% of time).
Yes	The time to cumulate 75% of suspended sediment transport is ±> 20% compared to its initial time (e.g., reduction from 20% to 16% of time).

9. Channel narrowing (R 1.1A)

Reduction in (bankfull) channel width (average on % of reach length) by comparing the present vs. reference condition is assessed (change in %). Rivers' bankfull channel width (and corresponding depth) might considerably be changed due to direct human interventions (e.g. river regulation, channel cut-off, etc.) and self-adjustment caused by alterations of the driving variables (water and sediment input). This problem also refers to the lack of morphological spatial demand: The Minimum Morphological Spatial Demand (MMSD) defines the space rivers need to naturally change during floods. Keeping this area free of development reduces flood damage, supports natural river processes, and improves flood management by allowing rivers to widen and shift within designated zones.

***Problem criteria* - Channel narrowing (R 1.1A)**	
No	No or slight average narrowing (≤ 10%) of the channel compared to near-natural channel width.
Yes	Average narrowing (> 10%) of the channel compared to near-natural channel width.

10. Excessive channel widening (R 1.1B)

Excessive increase in average (bankfull) channel width by comparing the present vs. reference condition is assessed (change in %). Excess in channel width considers direct artificial interventions increasing the natural (reference) channel width (e.g. construction of reservoirs or regulation of river channels to increase capacity). This parameter includes cases of water surface widening due to rise in water level such as cases of impoundment, dams, etc. without artificial channel widening (e.g. iron gate gorge or run-off river plants without channel widening). This parameter does not consider cases of restoration measures, where channel widths are not widened beyond the natural (reference) channel width.

***Problem criteria* - Excessive channel widening (R 1.1B)**	
No	No or slight average width increase (≤10%) compared to near-natural channel width.
Yes	Average width increase (>10%) compared to near-natural channel width.

11. Channel straightening (R 1.2)

River course adjustments aiming at straightening are assessed by evaluating the shortening of reference channel length in comparison with the present state. For this parameter the evidence of river course adjustments aiming at straightening (also considers case of channel re-routing which is a shift of channel course and associated length reduction) are assessed. Straightening only due to direct artificial interventions are considered. For rivers with more active channels (anabranching, braiding) the river course along the main (e.g. navigable) channel is considered.

***Problem criteria* - Channel straightening (R 1.2)**	
No	Not or slightly straightened river course: shortening ≤ 10% of the channel length;
Yes	Straightened river course: shortening > 10% of the channel length;

12. Floodplain disconnection (R 1.3)

Presence of longitudinal structures (embankments, artificial levees, flood dikes, infrastructure (e.g. roads) – Figure 24) higher than the floodplain level which interrupt hydrological connectivity and natural flooding of the riverine areas is assessed. Floodplain disconnection by longitudinal structures disrupts the natural hydromorphological processes of rivers, leading to reduced floodplain inundation, ecological degradation, increased flood risks, and reduced ecosystem functionality.

***Problem criteria* - Floodplain disconnection (R 1.3)**	
No	Reach length accompanied by longitudinal structures summed up on both banks is ≤ 10%.
Yes	Reach length accompanied by longitudinal structures summed up on both banks is > 10%.

13. Sediment extraction (e.g. commercial dredging, maintenance dredging) (R 2.1)

Occurrence of sediment dredging and its intensity, that is the % of historically transported sediments dredged in the past (from 20 to 100 years ago) in the reach is assessed. Sediment dredging (extraction of bed sediments out of the river) represents a great anthropogenic problem on the morphology of rivers (Figure 25). In the past commercial dredging leading to river bed degradation has been responsible for widespread negative consequences for rivers. Besides commercial dredging, maintenance dredging to maintain required fairway depth is also considered within this problem if dredged sediments are extracted from the river system.

***Problem criteria* - Sediment extraction (e.g. commercial dredging, maintenance dredging) (R 2.1)**	
No	Sediment dredging has never been done in the reach **or** Sediment dredging in the reach in the past (from 20 to 100 years ago) ≤ 10% of sediment transport in the past **and** no sediment dredging in the reach at present moment nor within the last 20 years.
Yes	Sediment dredging in the reach in the past (from 20 to 100 years ago) > 10% of sediment transport in the past **or** sediment dredging in the reach at present moment or within the last 20 years.

14. Bank reinforcement (R 2.2)

Presence of artificial bank protection of various types - hard bank reinforcement (concrete walls, rip rap, gabions, etc.) and soft reinforcement (bio-engineering solutions etc.) is assessed. Bank protection along the bank lines is considered being a limiting factor for lateral connectivity, channel natural evolution and source of sediments. Percentage of the total length of the banks is assessed. In case of anabranching rivers – total length is the sum of both bank lengths for each anabranch.

***Problem criteria* - Bank reinforcement (R 2.2)**	
No	Bank reinforcement on ≤ 10% of total length of the banks.
Yes	Bank reinforcement on > 10% of total length of the banks.

15. Transversal barriers in the reach (R 2.3)

Presence and density of transversal barriers with sediment trapping effect, ponding effect and/or disrupting longitudinal sediment connectivity within the reach are assessed:

- Barriers with low trapping effect such as consolidation weirs, crossing structures (culverts or bridge piers) and ramps
- Barriers with high trapping effect such as small dams or abstraction weirs/sluicing structures or retention weirs.
- Major barriers which are on the downstream border of the reach (=delineation criteria) are also assessed here.

***Problem criteria* - Transversal barriers in the reach (R 2.3)**	
No	No barriers in the reach **or** barriers with low sediment trapping effect in the reach.
Yes	Presence of barriers with high trapping effect in the reach **or** presence of major barriers on the downstream end of the reach.

16. Detrimental impacts of groynes (R 2.4)

Detrimental impacts of groynes refer to the negative effects that these river engineering structures can have on river morphology, sediment transport, and ecology. While groynes are designed to stabilize riverbanks and maintain navigation channels, they can alter flow patterns, cause local erosion or sediment accumulation, disrupt natural habitats, and reduce connectivity of aquatic and riparian ecosystems. Over time, these impacts may increase maintenance needs and reduce the ecological health of the river system.

***Problem criteria* - Detrimental impacts of groynes (R 2.4)**	
No	No groynes in the channel in the river reach **or** At least one groyne in the channel or groyne field at one river bank on ≤ 10% of the reach length.
Yes	One-side or both-sides of the channel with groyne fields (> 30%) of the reach length.

17. Bed reinforcement (R 2.5)

Presence of river bed reinforcement with impermeable or permeable surface as % of the reach length is assessed. Bed reinforcement mitigates river bed erosion and causes significant change in channel morphology: loss of vertical river bed connectivity, loss of habitats,

disruption of groundwater interaction (in case of impermeable bed reinforcement). Bed reinforcements predominately occur on mountain and foothill streams.

***Problem criteria* - Bed reinforcement (R 2.5)**	
No	Riverbed reinforcement on ≤ 10% of the reach length.
Yes	Riverbed reinforcement on > 10% of the reach length.

18. Impoundment (R 3.1)

Influence of artificial structures in the channel on water level regime and flow dynamics is assessed by the presence of impoundment in the reach (% of length of reach). This parameter should consider (the sum of) smaller impoundments in the reach caused by transversal barriers in the reach. For greater impoundments which are delineation criteria, the score is automatically 5 because the impoundment covers the total length of the assessed reach. Natural conditions of lowland rivers can exhibit low flow velocities or higher water depths etc. which are not considered as impoundment in this assessment.

***Problem criteria* - Impoundment (R 3.1)**	
No	Impoundment ≤ 5% of the reach length.
Yes	Impoundment > 5% of the reach length.

19. Water abstraction (R 3.2)

Water abstraction of the mean flow (Qmean) is assessed (% of Qmean). Abstraction for various purposes such as irrigation, derivation for HPPs, factories (water use, cooling, etc.) are considered here. Water abstraction disrupts the natural flow regime of rivers, leading to changes in sediment transport, habitat degradation, and reduced connectivity between the river and its floodplain. These changes can have cascading effects on the hydromorphological and ecological health of the river system.

***Problem criteria* - Water abstraction (R 3.2)**	
No	No abstractions or abstractions in the reach with sum of abstracted discharges < 1% of Q_{mean}.
Yes	One or more abstractions in the reach with sum of abstracted discharges < 1% of Q_{mean}.

20. Hydropeaking or artificial water level fluctuation (R 3.3)

Rate of water level fluctuations (cm/hour) or rapid in-/decrease of flow within the day are assessed. Operation of hydropower plants can result in artificial water level fluctuations. A significant case of water level fluctuation is related to hydropeaking. Hydropeaking is defined as discontinuous release of turbined water mainly due to peaks of energy demand, causing rapid artificial flow fluctuations into rivers downstream hydropower plants of reservoirs.

***Problem criteria* - Hydropeaking or artificial water level fluctuation (R 3.3)**	
No	Water level fluctuation ±> 5 cm/hour **or** rapid flow rate change, i.e. flow doubled or halved during ≤ 5% of time within the day.
Yes	Water level fluctuation ±> 5 cm/hour **or** rapid flow rate change, i.e. flow doubled or halved during > 5% of time within the day.

21. Channel aggradation (R 4.1A)

Change in river bed level is assessed by evaluation of sediment aggradation comparing present and past conditions. Natural conditions reflect dynamic morphological development including degradation and aggradation of sediments within the cross-sectional profile. This problem further involves sedimentation that impacts biota habitats and raises flood risk as bed levels increase.

***Problem criteria* - Channel aggradation (R 4.1A)**	
No	Slight change in sediment budget in the reach with prevailing aggradation and riverbed change ≤ + 1 cm/year.
Yes	Change in sediment budget in the reach with prevailing aggradation and riverbed change > + 1 cm/year.

22. Channel degradation (R 4.1B)

Change in river -bed level is assessed by long-term evaluation of degradation of the river bed. Channel degradation considers changes in sediment budget in the cross-sectional profile. Degradation can be caused by deficit in sediment supply, increased sediment transport capacity, lowering of bed level downstream, etc. River bed level change should be evaluated considering at least 10-year period between measurements – long term data recommended.

***Problem criteria* - Channel degradation (R 4.1B)**	
No	Slight change in sediment budget in the reach with prevailing degradation and riverbed change ≤ - 0,5 cm/year.
Yes	Change in sediment budget in the reach with prevailing degradation and riverbed change > - 0,5 cm/year.

23. Detrimental change of river morphological type (R 5.1)

Changes in river types comparing the present and reference condition is assessed as % of the reach length. Morphological river types can change due to shifts in sediment supply, water discharge, human interventions, vegetation dynamics, and broader environmental or climatic factors. These changes reflect the dynamic interplay between hydrology, sediment transport, and channel form. Detrimental changes or consequences can be ecological degradation, increased flood risks, and reduced functionality of river systems.

***Problem criteria* - Detrimental change of river morphological type (R 5.1)**	
No	Changes in river type are present < 10% of the reach length.
Yes	Changes in river type are present > 10% of the reach length.

24. Absence of an active floodplain (R 5.2)

<u>R 5.2 A: Remaining floodplain area</u>

This parameter assesses the remaining floodplain area by comparing the area of active floodplains of present and historical times. Active floodplains within this parameter refer to hydraulically active floodplains, which are defined as all areas that are still flooded during a HQ100 flood event.

***Problem criteria* - Absence of an active floodplain (R 5.2)**	
No	≥ 70% of the historical floodplain area is remaining at present times.

Yes	< 70% of the historical floodplain area is remaining at present times.

25. No typical fluvial landforms in the active floodplain (R 5.3)

The hydrological connectivity of the floodplain to the river channel is assessed by evaluation of presence of typical fluvial landforms in the active floodplain. Fluvial landforms in the active floodplain such as oxbow lakes, secondary channels, abandoned channels, relicts, wetlands, swamps, islands etc. are expected for individual river types. An active (or modern) floodplain is related to its connection to the river channel and its floodplain functionality (i.e. flooding is expected).

***Problem criteria* - No typical fluvial landforms in the active floodplain (R 5.3)**	
No	Natural or near-natural presence of fluvial landforms which are part of the active river system – undisturbed conditions without anthropogenic alteration **or** Presence of fluvial landforms which are partly or totally disconnected from the main channel, but are part of the active floodplain with potential for their reconnection.
Yes	Presence of fluvial landforms which are totally disconnected from the main channel, but are part of the active floodplain with potential for their reconnection **or** Limited occurrence of fluvial landforms or their relicts of high degradation, with low potential of their reconnection **or** Absence of typical fluvial landforms.

26. Limited cross-sectional profile variability (R 6.1)

Changes in the channel cross-section heterogeneity (= natural variability in channel depth at bankfull stage) are assessed. The assessment evaluates the degree of homogeneity of cross-sections on the reach length, which indicates a corresponding lack in cross-section heterogeneity. River channels cross-sections can considerably change from natural heterogeneity, because responses to alterations of the driving variables (water and sediment input) or direct artificial interventions such as river course adjustment or reinforcement of bank and river bed etc. However, the presence of bank protections is not sufficient for evaluating alteration of channel cross sectional variability. These structures over time can become morphologically masked by vegetation and sediments, and therefore be characterized by near-natural cross-section variability.

***Problem criteria* - Limited cross-sectional profile variability (R 6.1)**	
No	Artificial alteration of the channel shape (homogeneous cross-section profile) on ≤ 10% of the reach length.
Yes	Artificial alteration of the channel shape (homogeneous cross-section profile) on > 10% of the reach length.

27. Change of channel slope (R 6.2)

Change in average channel slope by comparing the present vs. past condition is assessed (change in%). Rivers' channel slope might considerably change or self-adjust due to alterations of the driving variables (water and sediment input), alterations of downstream boundaries such as lowering of bed level or water level increase but also in response to direct human interventions (e.g. channel straightening) in respective reach.

***Problem criteria* - Change of channel slope (R 6.2)**	
No	Slight average slope change (≤ 5%) of the channel.

Yes	Average slope change (> 5%) of the channel.

28. Detrimental change bed substrate grain size (R 6.3)

Change in median bed substrate grain sizes (D50), coarsening/fining, compared to undisturbed conditions is assessed. Change of the median grain size (D50) within the grain size distribution could indicate an imbalance between sediment supply and sediment transport capacity. Change in river bed substrate grain sizes, e.g., shift in grain size distribution (D50) towards coarsening could be an expected response change affected by upstream sediment barriers, increased sediment transport capacity, etc. Sampling of bed material D50 of present and undisturbed conditions (if available) are to be compared.

***Problem criteria* - Detrimental change bed substrate grain size (R 6.3)**	
No	Slight change in bed substrate D_{50} (≤ 20%) compared to undisturbed conditions.
Yes	Change in bed substrate D_{50} (> 20%) compared to undisturbed conditions.

29. River bank erosion (R 6.4)

Unprotected riverbanks can experience erosion during floods or from waves caused by passing ships. Persistent erosion or mass bank failure becomes a long-term problem. This parameter evaluates the presence (%) and distribution of bank erosion along the length of the reach. The parameter assesses whether natural bank erosion processes are present or whether the lateral evolution of the banks is lacking due to bank protection, decrease of bed slope, reduction of stream energy etc. The erosion of river banks (and, where applicable, hillslopes) as a natural process related to lateral evolution of the river channel provides sediment supply and diverse habitats. On the other hand, extreme bank erosion with widespread bank failures can occur within the reach - caused by sediment deficit (river bed incision) or strong alteration of flow regime.

***Problem criteria* - River bank erosion (R 6.4)**	
No	Natural lateral evolution of river banks under unaltered conditions **or** bank erosion present at > 30% of the reach length.
Yes	Bank erosion present at ≤ 20% of the reach length **or** bank erosion present at ≤ 10% of the reach length **or** evidence of eroding banks by mass failures (≤ 30% of the reach) **or** total absence of naturally eroding riverbanks in the reach **or** widespread presence of eroding banks by mass failures (> 30% of the reach).

30. Absence of riparian vegetation (R 7.1)

Length of riparian vegetation is assessed (% of length of both river banks). In this parameter the riparian vegetation (strip) is defined as vegetation on top of the bank plus vegetation within the bankfull channel (width ≥ 5 m). Riparian vegetation refers to its connectivity with the channel, where plants grow along the banks of the river, including trees, shrubs, and grasses having hydromorphologically functioning (flow resistance, bank stabilization, wood recruitment, sediment trapping, etc.). In case of bank reinforcement, vegetation bordering protected (artificially reinforced) river banks is taken into consideration because it can still be flooded allowing it to provide flow resistance, supply wood, and trap sediment.

***Problem criteria* - Absence of riparian vegetation (R 7.1)**

No	Riparian vegetation is present on ≥ 70% of both banks.
Yes	Riparian vegetation is present on < 70% of both banks.

31. Absence of active floodplain vegetation (R 7.2)

The length and average width of vegetation on the active floodplain is assessed. Vegetation along both river sides is taken into account. In this parameter the length, which is the sum of both banks (% of both banks - analogous to parameter [R 1.3]) and width of vegetation (floodplain forest) on the active floodplain, here considered from the top of the bank towards the edge of the active floodplain, are assessed. An active (or modern) floodplain is related to its connection to the river channel and its floodplain functionality (i.e. flooding is expected).

***Problem criteria* - Absence of active floodplain vegetation (R 7.2)**	
No	Active floodplain vegetation is present on ≥ 70% of the floodplain area, which is defined ≥ 4B for single-thread and anabranching channels, or ≥ 2B for braiding and wandering channels.
Yes	Active floodplain vegetation is present on < 70% of the floodplain area, which is defined ≥ 4B for single-thread and anabranching channels, or ≥ 2B for braiding and wandering channels.

32. Damages/Inundation of objects (FR 1)

This problem describes the physical impacts on buildings, transport infrastructure, utilities, and other assets in the river corridor and floodplain caused by flooding or rapid water level changes, including direct inundation, structural damage, erosion, access disruption, and sediment deposition. Mobilization and transport of large volumes of fine sediments during high flows can intensify these effects, resulting in severe damage to buildings, infrastructure, and agricultural land.

Problem criteria	
No	No observed or documented inundation or damage to objects within the river corridor or floodplain.
Yes	Observed or documented inundation and/or physical damage to buildings, infrastructure, utilities, or agricultural land caused by flooding or rapid water level changes, including effects from erosion or sediment deposition.

33. Economic damages through production losses (FR 2)

This problem refers to economic losses from reduced production caused by both floods and droughts. Floods and hydromorphological disturbances can interrupt industrial, commercial, agricultural, and service activities in river corridors and floodplains, leading to downtime, reduced output, and revenue losses, especially in exposed urban and industrial areas. Drought conditions, on the other hand, limit water availability, reducing hydropower generation, restricting navigation, and disrupting economic activities. Together, these effects lower productivity, increase costs, and impact supply chains.

Problem criteria	
No	No observed or documented inundation or damage to objects within the river corridor or floodplain.
Yes	Observed or documented economic losses due to reduced production caused by floods, droughts, or hydromorphological disturbances, including disruptions to industrial,

	commercial, agricultural, or service activities (e.g. downtime, reduced output, navigation or hydropower limitations).

34. Risk of dike failure (FR 4)

This problem refers to the likelihood of flood protection embankments overtopping, breaching or being structurally compromised during high-flow events, causing rapid inundation of protected areas. Confining rivers with nearby flood defense structures reduce lateral retention and increase shear stress and incision risk, which can elevate hydraulic loads on dikes during floods.

Problem criteria	
No	No observed or identified risk of dike failure or structural instability under current or expected flood conditions.
Yes	Observed or identified risk of dike overtopping, breaching, or structural instability during high-flow events, including increased hydraulic loads due to river confinement, reduced retention capacity, or erosion processes.

35. Channel blockage (FR 6)

Channel blockage in the context of floods refers to temporary obstructions of the flow cross-section when debris (e.g., large wood, sediment, vegetation) accumulates at constrictions such as bridges, culverts, or other transversal structures, narrowing the channel, creating backwater, and raising upstream water levels and flood risk.

Problem criteria	
No	No observed or identified occurrence or risk of channel blockage affecting flow conditions during flood events.
Yes	Observed or identified occurrence or risk of channel blockage during high-flow events due to accumulation of debris (e.g. large wood, sediment, vegetation) at bridges, culverts, or other structures, leading to flow obstruction, backwater effects, or increased upstream flood levels.

36. High maintenance costs (FR 8)

High maintenance costs arise from the need to regularly inspect, operate, and repair infrastructure to manage flood and drought risks and ensure safety and functionality. This includes hydraulic and flood retention structures such as reservoirs, polders, and basins. Additional costs stem from ecological maintenance (e.g., habitat restoration, bank stabilization), sustaining hydropower functionality (e.g., turbine inspections, sediment removal), and navigation (e.g., dredging, sediment reinsertion, channel marking). These activities require ongoing financial investment to preserve system performance and reduce environmental and economic risks.

Problem criteria	
No	Maintenance costs remain within typical/expected ranges (e.g. <2% of asset value, no unusual increase, no extraordinary interventions)
Yes	High maintenance costs (Yes) if at least one applies: • Annual maintenance costs exceed 5% of the asset replacement value, or • Annual costs show a sustained increase (>20% over 5 years), or • Maintenance costs significantly exceed regional/sectoral benchmarks, or

	• Extraordinary measures (e.g. frequent dredging, repeated repairs) are required beyond routine maintenance

37. Decrease of river water level (DR 2)

This parameter describes reductions in river water levels, caused by water withdrawals or changes in the river system. Lower flows can affect hydropower, navigation, drinking water supply, agriculture, recreation, and aquatic ecosystems. The decrease is assessed by comparing current river water levels (e.g., the past 5–10 years) with a long-term reference period (e.g., 20–30 years). If no measured or modeled data are available, the decrease of river water level should be assessed qualitatively based on expert judgment, historical observations, local knowledge, or reports of impacts on water uses and ecosystems.

Problem criteria	
No	Long-term river water level is ≤20% compared to the past period
Yes	Long-term river water level has decreased by >20% compared to the past period

38. Decrease of GW-level and recharge (DR 3)

This problem describes sustained drops in groundwater levels together with reduced replenishment of aquifers, commonly caused by excessive water abstraction and by alterations of the hydrological regime that weaken surface–groundwater connections. Climatic changes that increase evaporation and reduce groundwater storage and recharge further contribute to prolonged declines in groundwater heads. Use long-term historical groundwater levels and recharge rates (e.g., past 20–30 years) as baseline. Determine current conditions based on measured groundwater levels, spring discharge, or modeled recharge data (e.g., over the past 5–10 years). If no data is available, this parameter is assessed qualitatively based on expert judgment, observations, and available reports (e.g., observed lowering of wells, drying of springs, or ecosystem stress).

Problem criteria	
No	Decreases are ≤20%, with negligible impacts on water uses or ecosystems.
Yes	Groundwater levels or recharge have decreased by >20% compared to the reference period, leading to noticeable impacts on water availability or ecosystems.

39. Drying out of side arms (DR 5)

This problem refers to the drying out or loss of flow in side arms and oxbows during low-flow periods, caused by reduced connection to the main river. A key reason is the build-up of fine sediment at the inlets, which gradually blocks the connection and isolates these channels. In slow-flowing backwater areas, low velocities and high sediment loads lead to further sediment deposition, which reduces water flow and speeds up disconnection and drying. This parameter is assessed qualitatively based on expert judgment, observations, and available reports.

40. Ecological status worse than good (EBR 1)

This problem refers to rivers not reaching good ecological status, as required by the Water Framework Directive. Ecological status is based mainly on biological quality, supported by

chemical and physical conditions as well as hydromorphological characteristics such as flow, water depth and riverbed structure.

41. Ecological potential worse than good (EBR 2)

This parameter, defined under the Water Framework Directive, refers to water bodies that do not achieve good ecological potential. Ecological potential describes how close a heavily modified or artificial water body is to its best possible ecological condition, given its hydromorphological alterations. Maximum ecological potential represents the highest condition that can be reached with all feasible improvement measures that do not significantly affect water use or ecology. Good ecological potential means only a small deviation from this state. If the deviation is greater, the water body is classified as moderate, poor, or bad ecological potential, indicating the need for further restoration measures.

42. Excess nutrients (EBR 3)

This problem refers to high inputs of nitrogen and phosphorus into water bodies, which cause excessive growth of algae and aquatic plants and lead to eutrophication. Nutrients mainly enter through diffuse pathways, with nitrogen transported via subsurface flow and phosphorus through soil erosion. Additional inputs come from urban wastewater and industrial discharges, especially where treatment is insufficient. Use water quality standards, background concentrations, or environmental quality standards (EQS) as baseline. Determine current conditions based on monitoring data for nitrogen (e.g., nitrate, ammonium) and phosphorus (e.g., total phosphorus, orthophosphate) in water or sediments. If no data is available, assessment is made qualitatively based on expert judgment, observations, or available reports (e.g., visible algal blooms, reported water quality issues, known nutrient sources).

Problem criteria	
No	Nutrient concentrations are within natural or regulatory thresholds, with no significant ecological impact.
Yes	Measured nutrient concentrations exceed environmental quality standards or background levels, indicating risk of eutrophication or ecological impacts.

43. Water temperature increase (EBR 4)

This parameter describes rises in river water temperature beyond natural or reference conditions, which can affect aquatic ecosystems, species survival, water quality, and ecosystem functioning. Increases may result from reduced flows, climate change, or thermal discharges from industrial or urban sources. Use long-term historical water temperature data for the river as baseline and compare it with existing monitoring data (daily, seasonal, or annual averages) or modeled temperature. If no data is available, assessment is made qualitatively based on expert judgment, observations, or available reports (e.g., fish mortality events, changes in species composition, observed warming trends).

44. Sediment contamination (EBR 6)

This problem refers to the accumulation of harmful substances in river sediments, including heavy metals and persistent organic pollutants (POPs). These substances can damage aquatic ecosystems and pose risks to biota and humans. Pollution stems from historical and current sources, such as industry, mining, urban runoff, and wastewater. Activities like flushing or natural floods can stir up these polluted sediments and spread contaminants if

not properly managed. This parameter is assessed qualitatively based on expert judgment, observations, and available reports (e.g. known pollution sources, historical contamination, discoloration).

Problem criteria	
No	Water temperature increases are ≤2 °C, with no significant ecological or water quality impacts.
Yes	Water temperature increases by >2 °C above reference conditions, causing measurable stress to aquatic ecosystems or water quality degradation

45. Continuity interruption - biota, fish (EBR 7)

This problem refers to the interruption of river continuity by structures such as dams, weirs, and ramps. These barriers hinder fish migration and limit access to relevant habitats and spawning grounds. As a result, fish populations become fragmented, especially affecting species that migrate over long distances along the Danube and its tributaries. If no information is available, this parameter is assessed qualitatively based on expert judgment, observations, and available reports.

Problem criteria	
No	The structure allows unrestricted passage for the entire typical fish fauna and all relevant age stages throughout the year.
Yes	Fish passage is possible for the entire typical fish fauna under favourable conditions, but only for species with good swimming ability or adult individuals under less favourable conditions.

46. Presence of macro and microplastics (EBR 11)

This problem refers to the occurrence of plastic debris (macroplastics) and small plastic particles (microplastics) in waters, sediments, and biota across the basin. Plastic pollution in freshwater systems is widespread and particles are ingested by a broad range of organisms, raising concerns for organism health and the integrity of aquatic food webs. The best way of reducing litter and micro plastics entering the water systems is reduction at the sources. This parameter is assessed qualitatively based on expert judgment, observations (e.g. visible litter), and available reports on plastic pollution sources or accumulation zones.

47. Presence of hazardous substances (EBR 12)

Hazardous substances pollution refers to contamination by the priority substances listed in WFD Annex X and other specific pollutants in WFD Annex VIII that are toxic, persistent, accumulative and have local or regional relevance. These include inorganic and organic micro-pollutants such as heavy metals, arsenic, cyanides, and oil. Emissions arise from both point (e.g. wastewater treatment plants) and diffuse sources. Diffuse pathways are substance-specific, with surface runoff, sediment transport, and groundwater flow being the principal routes. This parameter is assessed qualitatively based on expert judgment, observations, and available reports (e.g. known pollution sources, past contamination events, visible impacts).

48. Insufficient ecological flow (SHR 4)

This problem occurs where the residual or released flow downstream of abstractions or dams is too low to sustain aquatic ecosystems, indicating a human-altered hydrological regime that degrades ecological integrity. Reduced flows diminish habitat quantity and quality, disrupt longitudinal and lateral connectivity, and impair aquatic biota, with documented impacts on invertebrate and fish populations, including migration and spawning success. Short-term discharge fluctuations from hydropeaking further stress communities by rapidly altering water depths and velocities. If no ecological flow requirements are available, this parameter is assessed qualitatively based on expert judgment, observations, and available reports (e.g. observed habitat degradation, fish stress, drying sections, or water quality issues).

Problem criteria	
No	River flow is matching ecological flow requirements, with no significant ecological impacts
Yes	River flow is below ecological flow requirements or natural reference conditions during relevant periods, leading to ecological stress or habitat degradation.

49. Aging and inefficient hydropower (SHR 5)

This problem refers to older HPP whose equipment and layouts underperform in energy efficiency and operational reliability, often lagging behind current environmental standards and practices. Upgrading existing plants is highlighted as a key pathway to improve both power generation and environmental performance (e.g., functional fish passage, habitat restoration, sediment management, and environmental flows). This parameter is assessed qualitatively based on expert judgment, observations, and available reports.

50. Loss of reservoir volume due to sedimentation (SHR 6)

Reservoirs progressively lose storage as sediments accumulate, reducing operability and flood-control capacity. Such trapping also propagates sediment deficits downstream, contributing to riverbed incision and habitat degradation below dams. To limit this issue, sediment can be routed during high flows by sluicing or flood-conditioned flushing to pass more load through impoundments and reduce deposits. Where routing alone is insufficient or retrofitting is not feasible, dredging with reinsertion downstream can recover reservoir volume. If no measured or modeled data are available, the problem should be assessed qualitatively based on expert judgment, historical observations, local knowledge, or reports of impacts on water uses and ecosystems.

Problem criteria	
No	Volume loss ≤20% with no significant impact on reservoir function
Yes	Loss of reservoir volume >20% compared to original capacity or significant reduction in functionality

51. Endangered proper hydropower operation due to sedimentation at the dam (SHR 7)

This problem refers to sediment accumulation at dams and intakes, which impair operability by reducing conveyance and clogging structures. When bedload and fine sediments accumulate in the impoundment and cannot pass, they often must be removed and

reintroduced downstream to maintain continuity and plant functionality. This parameter is assessed qualitatively based on expert judgment, observations, and available reports.

52. Fairway depth less than 2.5 m (SNR 1)

This problem refers to insufficient navigation depth (below 2.5 m) along designated fairways, which restricts safe and reliable ship passage, particularly during low-flow periods. Reduced depth increases the risk of grounding, limits cargo capacity, and disrupts schedules. The primary cause is enhanced sediment deposition in areas of decreased flow velocity (e.g., impounded reaches), leading to progressive loss of depth. Additional factors include changes in channel morphology, tributary sediment inputs, and inadequate maintenance dredging.

53. Fairway width less than 100 - 180 m (SNR 2)

This problem refers to insufficient navigation corridor width (less than 100–180 m) along designated fairways, which constrains safe ship passage and overtaking. A common cause is local sedimentation along channel margins and bars in zones of reduced flow velocity (e.g., impounded sections), which intrudes into the navigable cross section and narrows the usable fairway.

54. Unfavourable flow conditions at ship locks (SNR 3)

This problem refers to unfavourable hydraulic conditions in lock approaches that reduce navigation safety and reliability by impeding safe ship manoeuvring and passage. The creation of currents, turbulence and locally high velocities linked to operational releases (e.g., bottom outlets and gates) is a typical cause that needs to be carefully coordinated with navigation to prevent affecting ship traffic. This parameter is assessed qualitatively based on expert judgment, observations, and available reports.

55. Unfavourable flow conditions at junctions (e.g. side arms) (SNR 4)

This problem refers to unfavourable hydraulic conditions, such as extremely turbulent zones, transverse flows and varying water levels, at river junctions (e.g. side arms). These conditions hinder safe ship approach, passage and overtaking, thereby decreasing navigation safety and reliability. This parameter is assessed qualitatively based on expert judgment, observations, and available reports.

56. Height under non-openable bridged less than 5.25 m (SNR 5)

This problem refers to insufficient clearance under non-openable bridges, which restrict or prevent safe ship passage and reduce navigation reliability. A common cause is elevated water levels in the reach due to impoundment/backwater effects from transversal structures.